

Chapter 6

6.1. Try to find the unilateral Laplace transform and convergent domain of the periodic signal shown in figure 6-1.

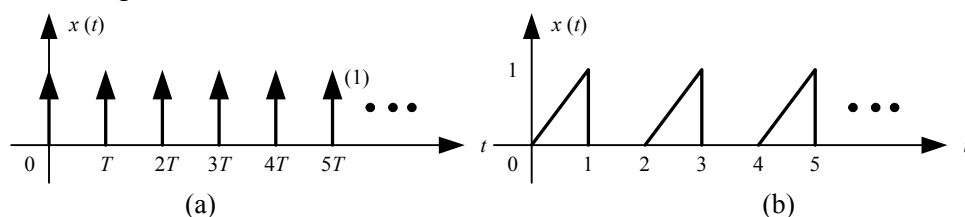


Figure.6-1

6.2. We try to find the Laplace transformation of the following functions by using the property of Laplace transform.

(1) $\frac{d}{dt}[e^{-2t}u(t)]$

(2) $\delta(3t-2)$

(3) $te^{\lambda t} \cos(\omega_0 t)u(t)$

(4) $\int_0^t e^{-\lambda(t-\tau)} \cos(\omega_0 \tau) d\tau, t \geq 0$

6.3. We know $\mathcal{L}\{x(t)\} = X(s) = \frac{1}{s+1}$, $\text{Re}\{s\} > -1$. Try to find the unilateral Laplace transform of

the following signals using the property of Laplace transform.

(1) $x_1(t) = x(t-1)$

(2) $x_2(t) = x(2t)$

(3) $x_3(t) = x(2t-2)$

(4) $x_4(t) = e^{-t}x(t)$

(5) $x_5(t) = x'(t)$

(6) $x_6(t) = tx(t)$

(7) $x_7(t) = x(t) * x(2t)$

(8) $x_8(t) = x(2t) * x(2t)$

6.4. Try to find the initial value $x(0^+)$ and final value $x(\infty)$ of the following $X(s)$.

(1) $X(s) = \frac{2s+1}{s(s+2)}$, $\text{Re}\{s\} > 0$

(2) $X(s) = \frac{3s+2}{(s+1)(s+2)(s+3)}$, $\text{Re}\{s\} > -1$

(3) $X(s) = \frac{s^2+2s+3}{s(s+2)(s^2+4)}$, $\text{Re}\{s\} > 0$

(4) $X(s) = \frac{2s^2+1}{s(s+2)}$, $\text{Re}\{s\} > 0$

6.5. Try to find the following $X(s)$ unilateral Laplace inverse transformation using partial fraction method.

(1) $X(s) = \frac{3s+2}{s^2+4s+3}$, $\text{Re}\{s\} > -1$

(2) $X(s) = \frac{2s+1}{s(s+1)(s+3)}$, $\text{Re}\{s\} > 0$

(3) $X(s) = \frac{5s+13}{s(s^2+4s+13)}$, $\text{Re}\{s\} > 0$

(4) $X(s) = \frac{1}{(s+1)(s+2)^2}$, $\text{Re}\{s\} > -1$

(5) $X(s) = \frac{3s}{(s^2+1)(s^2+4)}$, $\text{Re}\{s\} > 0$

(6) $X(s) = \frac{s}{s^3+2s^2+9s+18}$, $\text{Re}\{s\} > 0$

6.6. Try to find the following $X(s)$ unilateral Laplace inverse transformation using partial fraction method.

(1) $X(s) = \frac{s^2+7s}{s^2+6s+8}$, $\text{Re}\{s\} > -2$

(2) $X(s) = \frac{s^2+6s+6}{s+6}$, $\text{Re}\{s\} > -6$

(3) $X(s) = \frac{1+2e^{-4s}}{(s+1)(s+2)}$, $\text{Re}\{s\} > -1$

(4) $X(s) = \frac{se^{-2s}}{(s+1)(s+2)^2}$, $\text{Re}\{s\} > -1$

6.7. The figure 6-2 shows simply low pass, high pass, band pass and band stop system respectively.

Obtain the system functions of each system and plot the amplitude response of the system. And $R=1\Omega$, $C=1F$, $L=1H$

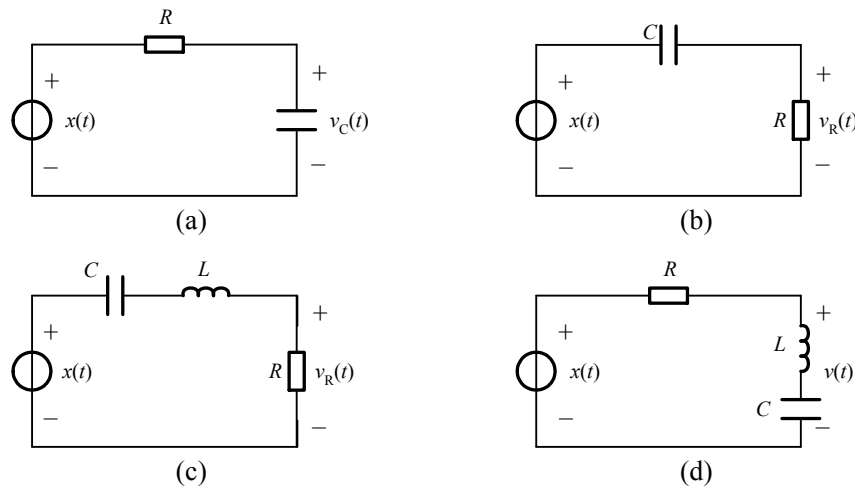


Figure.6-2

6.8. According to the zero and pole of the system shown in figure 6-3, try to plot the amplitude response curves of each diagram, assuming the gain constant of all systems is $K=1$.

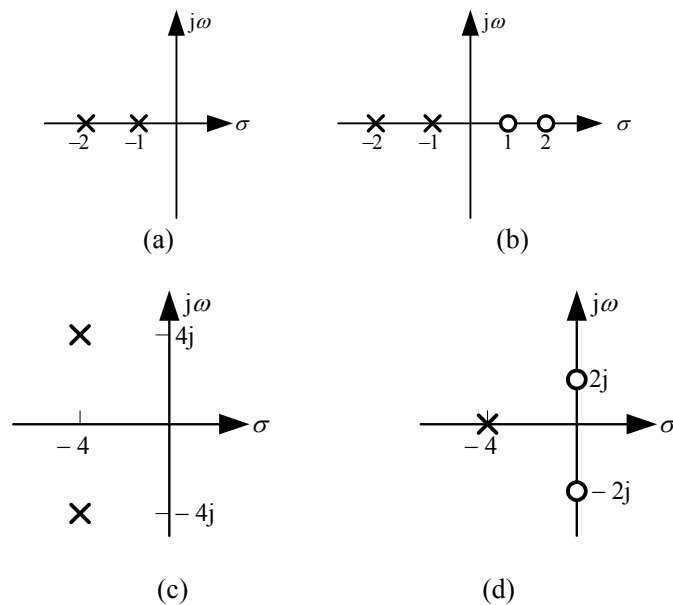


Figure.6-3

6.9. The zero state response of a continuous causality LTI system under the action of $\delta'(t)$ is

$$y_{zs}(t) = 3e^{-2t}u(t), \text{ try to determine by s domain.}$$

(1) Impulse response $h(t)$ of the system

(2) Zero state response generated by the system for input $x(t) = 2[u(t) - u(t-2)]$

6.10. The differential equation describing a causality continuous time LTI system is

$$y''(t) + 7y'(t) + 10y(t) = 2x'(t) + 3x(t)$$

And $x(t) = e^{-2t}u(t)$, $y(0^-) = 1$, $y'(0^-) = 1$, try to determine by s domain.

(1) Zero input response $y_{zi}(t)$, zero state response $y_{zs}(t)$, complete response $y(t)$.

(2) The system function $H(s)$, the unit impulse response $h(t)$ and the stability of the system.

(3) Plot the direct simulation block diagram of the system

6.11. Draw the following direct block diagram of the following causal continuous time LTI system.

$$(1) H(s) = \frac{5(s+1)}{(s+2)(s+5)}$$

$$(2) H(s) = \frac{s^2 + 2s - 3}{(s+2)(s+5)}$$

$$(3) H(s) = \frac{s-3}{s(s+1)(s+2)}$$

$$(4) H(s) = \frac{2s-4}{(s^2-s+1)(s^2+2s+1)}$$

6.12. Try to find the system function of causality continuous time system shown in figure 6-4 and determine the system stable constant β .

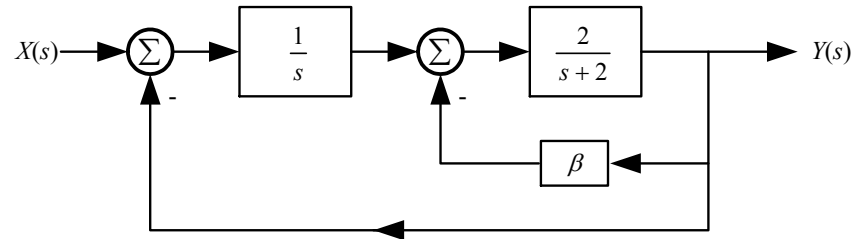


Figure.6-4